

Discovery of Length-7 Prime Chains under a 4-adic Collatz-like Map

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Abstract

We report the discovery of two prime chains of length 7 generated by the 4-adic Collatz-like function:

- If $n \equiv 1 \pmod{4}$: $f(n) = (5n - 1) / 4$
- If $n \equiv 3 \pmod{4}$: $f(n) = (5n + 1) / 4$
- Otherwise: terminate

Using a modular skipping method, we identified two such chains within the range $k \leq 1,600,000$, both starting from $n_0 = M \cdot k + 1$, where

$$M = 4^7 \times (3 \times 7 \times 11 \times 13 \times 17 \times 19 \times 23) = 365,515,358,208.$$

Definition

A 4-adic prime chain is a finite sequence of primes $(p_0, p_1, \dots, p_6)$ such that:

- $p_{i+1} = f(p_i)$
- All p_i are prime
- The chain ends when $f(p_k)$ is composite or even

Discovered Chains

Chain 1 ($k = 1,247,136$)

$$\begin{aligned} n_0 &= 455,847,361,774,092,289 \\ n_1 &= 569,809,202,217,615,361 \\ n_2 &= 712,261,502,772,019,201 \\ n_3 &= 890,326,878,465,024,001 \\ n_4 &= 1,112,908,598,081,280,001 \\ n_5 &= 1,391,135,747,601,600,001 \\ n_6 &= 1,738,919,684,502,000,001 \end{aligned}$$

Chain 2 ($k = 1,529,862$)

$$\begin{aligned} n_0 &= 559,188,056,938,807,297 \\ n_1 &= 698,985,071,173,509,121 \\ n_2 &= 873,731,338,966,886,401 \\ n_3 &= 1,092,164,173,708,608,001 \\ n_4 &= 1,365,205,217,135,760,001 \\ n_5 &= 1,706,506,521,419,700,001 \\ n_6 &= 2,133,133,151,774,625,001 \end{aligned}$$

All values were verified prime using deterministic Miller-Rabin testing.

Method

To avoid even-number traps and small prime divisors, we used a skipping algorithm based on modular constraints. The modulus

$$M = 4^7 \times \text{primorial}(23) = 365,515,358,208$$

ensures survival for 7 steps and avoids small factors. Candidates of the form $n_0 = M \cdot k \pm 1$ were tested using a parallelized Python script with `sympy.isprime`.

Conclusion

This study confirms the existence of 7-term prime chains under a 4-adic Collatz-like map. The skipping method enables efficient discovery and may extend to longer chains ($L \geq 8$).

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Code and data are available with the Zenodo submission.